

Minimizing Modality Gap from the Input Side: Your Speech LLM Can Be a Prosody-Aware Text LLM

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Abstract

Speech large language models (SLMs) are typically built from text large language model (TLM) checkpoints, yet they still suffer from a substantial modality gap. Prior work has mainly attempted to reduce this gap from the output side by making speech generation more text-like, but the gap remains. We argue that the key remaining bottleneck lies on the input side. We propose TextPro-SLM, an SLM that makes spoken input more closely resemble that of a prosody-aware text LLM. TextPro-SLM combines WhisperPro, a unified speech encoder that produces synchronized text tokens and prosody embeddings, with an LLM backbone trained to preserve the semantic capabilities of the original TLM while learning paralinguistic understanding. Experiments show that TextPro-SLM achieves the lowest modality gap among leading SLMs at both 3B and 7B scales, while also delivering strong overall performance on paralinguistic understanding tasks. These gains are achieved with only roughly 1,000 hours of LLM training audio, suggesting that reducing the modality gap from the input side is both effective and data-efficient.

1 Introduction

Speech Large Language Models (Speech LLMs, SLMs) have been extensively studied in recent years to enable seamless speech-based interaction [1–4]. Most SLMs are built by continually training from Text Large Language Model (TLM) checkpoints [5–10], with the goal of extending the strong reasoning and language understanding capabilities of TLMs to spoken input and output. However, despite this shared backbone, a major challenge remains: the **modality gap**—speech-based question answering (QA) performance often remains substantially worse than the text-based QA performance of the underlying TLM, limiting the practical usability of SLMs. For example, prior studies [11] report that GLM-4-Voice [12] suffers up to a 20% performance drop on several QA benchmarks.

Most existing studies attempt to bridge the modality gap from the **output side**. Early SLMs use the LLM backbone to generate only speech tokens, enabling fully end-to-end spoken interaction [13–15]. Later work improves performance by having the LLM first generate intermediate text tokens before emitting speech tokens [12, 16–18]. More recent approaches further decouple text generation from speech synthesis through a “thinker-talker” architecture, in which the LLM backbone generates only text tokens while a separate talker module produces speech conditioned on the thinker’s hidden states [19–23]. This line of progress reveals a clear pattern: the closer the generation process is to the interaction paradigm of a TLM, the better the resulting SLM tends to perform. Nevertheless, even these advances do not eliminate the modality gap.

This observation suggests that the remaining bottleneck may lie primarily on the **input side**. Under modern thinker-talker designs, the output side already resembles that of a TLM. The more fundamental difficulty is that, unlike a TLM, an SLM must infer both linguistic content and paralinguistic cues directly from speech representations, which are often not presented to the LLM in a form it can

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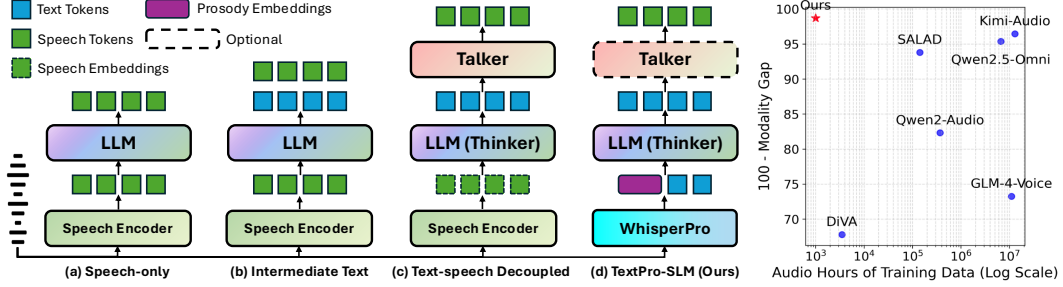


Figure 1: Left: Comparison of the architectural designs used by prior SLMs and our approach. Right: Comparison of modality gap versus the amount of audio training data across different approaches.

fully exploit [24, 25]. Motivated by this perspective, we propose to minimize the modality gap from the input side by making a Speech LLM behave as closely as possible to a **prosody-aware text LLM**. The central idea is simple: if the output side of modern SLMs is already converging toward text-style generation, then the input side should also provide the backbone with a representation that is both TLM-compatible and prosody-preserving. Rather than treating speech purely as an acoustic signal that must be compressed into a single latent stream, we explicitly separate and preserve two complementary aspects of spoken input: *what is said* and *how it is said*.

To achieve this, we propose **TextPro-SLM**, which consists of two key components. **1) Speech encoder.** We introduce **WhisperPro**, a unified **Whisper**-based speech encoder that produces two outputs: text tokens that capture utterance content and **Prosody** embeddings that encode paralinguistic information such as speaking style, emotion, and speaker timbre. WhisperPro is trained to support both accurate transcription and prosody-rich representation learning, so the downstream LLM receives a richer and more TLM-compatible view of the input speech. **2) LLM backbone.** We train the LLM backbone with two coupled objectives: preserving the semantic capabilities of the original TLM and learning to understand paralinguistics. Specifically, we retain the semantic intelligence of the underlying TLM through knowledge distillation, while jointly optimizing the model on paralinguistic-understanding tasks. In this way, both linguistic meaning and paralinguistic information can propagate through the model, yielding an end-to-end SLM that remains semantically strong while also interpreting how something is spoken. Figure 1 compares the architectural design of TextPro-SLM with prior solutions.

Experimental results show that our method not only achieves the lowest modality gap among the compared approaches but also attains competitive performance on paralinguistic-understanding tasks, while remaining extremely **data efficient**, requiring only roughly 1,000 hours of audio data for LLM training. We attribute this efficiency to the fact that our design stays closer to the original operating regime of the underlying TLM than prior SLM architectures, making adaptation from text to speech more direct and less data-intensive. Overall, our results suggest that an effective route toward stronger Speech LLMs is to turn them into prosody-aware text LLMs.

2 Related Work

2.1 Speech Large Language Models

SLMs aim to enable seamless spoken dialogue interactions [1]. A typical SLM consists of three main components: 1) a speech encoder that converts the input waveform into speech representations, typically either continuous speech embeddings [22, 23] or discrete speech tokens [26–28]; 2) an LLM backbone that processes these representations and generates outputs, which may be text tokens [23], speech tokens [14], or both [12]; and 3) a speech decoder that synthesizes the model output into spoken responses. Early SLMs are trained purely on speech data and typically represent speech as fully discrete tokens [13]. Later work shows that leveraging a pre-trained TLM as the backbone leads to substantially stronger performance on speech tasks [14]. Since then, building SLMs from TLM checkpoints has become standard practice, with much of the research focusing on explicit speech-text alignment leveraging Automatic Speech Recognition (ASR), Text-to-Speech Synthesis (TTS), or interleaved speech-text training data [12, 15, 29].

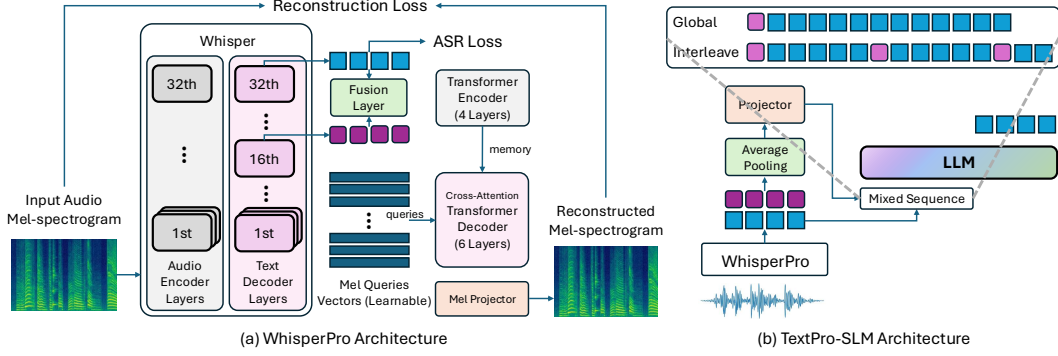


Figure 2: Model architecture of WhisperPro and TextPro-SLM.

Recent SLMs can be broadly grouped into two architectural paradigms according to the form of speech input provided to the LLM. **1) Discrete models:** the speech encoder produces discrete speech tokens, and the LLM consumes these tokens directly to generate text tokens, speech tokens, or both [12, 18]. **2) Continuous models:** the speech encoder produces continuous embeddings, which are then mapped into the LLM embedding space through a projector module. Models in this category often adopt a thinker-talker architecture, where the thinker generates text tokens and a separate talker produces speech conditioned on the thinker’s hidden states [22, 23]. Our approach differs from both paradigms by providing the LLM with text tokens together with prosody embeddings, rather than asking it to infer both semantic and prosodic information from a single speech representation. This design better preserves the operating regime of the underlying TLM and helps reduce the text-speech modality gap effectively.

2.2 Text-Speech Modality Gap

As discussed in Section 1, recent SLM architectures have narrowed the modality gap mainly by improving the output side. Here, we review work that explicitly studies this gap and work that attempts to reduce it, then clarify how our approach differs.

Some studies focus on understanding the gap itself. Hsu *et al.* [30] attribute it to the longer speech sequence and the sparsity of semantic information in speech. Xiang *et al.* [24] analyze the gap at the representation level and show that speech and text representations are similar in direction but differ in magnitude. Other studies aim to reduce the gap, mostly through improved training. SALAD [11] and Wang *et al.* [31] use Kullback-Leibler (KL) divergence to align the student’s output on speech input with the teacher’s output on text input. X-OPD [32] and CORD [33] extend this line with on-policy distillation, while Yang *et al.* [34] perform finer-grained layer-wise distillation. A separate line of work targets catastrophic forgetting during continual training on speech data: DeepOmni [35] uses a Mixture-of-Experts architecture [36] that separates text and speech experts, and Hsiao *et al.* [37] identify experience replay as the most effective continual-learning strategy [38]. However, these methods mainly improve optimization and training stability rather than the interaction paradigm itself, which we view as a more fundamental source of the modality gap.

3 Methodology

Our approach to minimizing the modality gap has two core components. Section 3.1 presents WhisperPro, our speech encoder, and Section 3.2 introduces the prosody-aware LLM backbone.

3.1 WhisperPro: An ASR Speech Encoder that Encodes Everything

We design a speech encoder that outputs both accurate transcripts and prosody-rich embeddings. Broadly, there are three ways to build such a model: starting from an ASR model and adding paralinguistic modeling, starting from a prosody-rich model such as a neural audio codec and adding transcription ability, or training a new model from scratch for both goals. We adopt the first strategy for two reasons. First, prior work suggests that SLMs rely more heavily on semantic information

than on fine-grained acoustic details [39, 40]. Second, strong ASR models are typically trained on much larger datasets than codec models. For instance, Whisper-large-v3 [41] is trained on 680,000 hours of audio, whereas a competitive codec model such as WavTokenizer [42] only requires as few as 585 hours. Starting from a strong ASR backbone therefore fits the downstream LLM well and requires significantly less training data. In practice, we choose Whisper-large-v3 [41] because of its strong transcription performance and robustness across diverse acoustic conditions.

The main challenge of this task is that ASR models are trained to prioritize lexical content, not prosody. To inject paralinguistic information into the representation space, we borrow the core idea behind codec models: preserve acoustic details through reconstruction. Concretely, we use Whisper as the encoder to produce both text tokens and token-aligned prosody embeddings, and we attach an additional decoder that reconstructs the input speech from these two outputs. This design encourages the encoder states to retain not only what was said, but also how it was said. Figure 2 illustrates the overall architecture of WhisperPro.

Architecture. The architecture of WhisperPro consists of two modules: Whisper-large-v3 and a mel-reconstructor. Let the input utterance be a waveform $\mathbf{x} \in \mathbb{R}^T$. We first follow Whisper’s standard preprocessing pipeline to obtain a log-mel spectrogram $\mathbf{M} \in \mathbb{R}^{F \times L}$. Whisper-large-v3 then maps $\mathbf{M}$ to both the transcript and the prosody embeddings. Specifically, Whisper autoregressively produces a text token sequence

$$\mathbf{y} = \text{Whisper}(\mathbf{M}) = (y_1, \dots, y_N). \quad (1)$$

Whisper-large-v3 consists of 32 Transformer encoder layers and 32 Transformer decoder layers. We take the hidden states of each text token from the 16-th Whisper decoder layer as token-aligned prosody embeddings, denoted as

$$\mathbf{P} = \text{WhisperDec}^{(16)}(\mathbf{M}, \mathbf{y}) = (p_1, \dots, p_N) \in \mathbb{R}^{N \times d}. \quad (2)$$

Because these states are aligned with the generated text tokens, WhisperPro represents speech as two synchronized streams, $\{(y_i, \mathbf{p}_i)\}_{i=1}^N$, capturing *what* was said and *how* it was said. To force the model to preserve prosodic information, we attach a mel-reconstructor that predicts the original mel spectrogram from the transcript-prosody pairs. At the overview level, we denote this module as

$$\hat{\mathbf{M}} = \text{MelReconstructor}(\mathbf{y}, \mathbf{P}). \quad (3)$$

Concretely, the text tokens and prosody embeddings are mapped into a shared space, fused token-wise, contextualized by a Transformer encoder, and decoded into frame-level acoustic features by a Transformer decoder with learnable frame queries. The detailed architecture of the mel-reconstructor is presented in Appendix B. This reconstruction objective encourages the decoder states to retain information not only for transcription, but also for recovering the acoustic realization of the utterance.

During training, WhisperPro is optimized with a multi-task objective consisting of an ASR loss and a Mel reconstruction loss. Let $\mathbf{y}^* = (y_1^*, \dots, y_N^*)$ denote the ground-truth transcription. The ASR objective is the average token-level cross-entropy:

$$\mathcal{L}_{\text{ASR}} = -\frac{1}{N} \sum_{i=1}^N \log p(y_i^* | y_{<i}^*, \mathbf{M}), \quad (4)$$

and the mel-reconstructor is trained with a mean squared error loss between the reconstructed and original Mel spectrograms:

$$\mathcal{L}_{\text{mel}} = \frac{1}{FL} \left\| \hat{\mathbf{M}} - \mathbf{M} \right\|_2^2. \quad (5)$$

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{ASR}} + \lambda \mathcal{L}_{\text{mel}}, \quad (6)$$

where λ is a hyperparameter and is empirically set to 1.0.

Training. We use the 960-hour LibriSpeech training split [43] as our primary training corpus. Because the original LibriSpeech transcripts are written in uppercase and do not include punctuation, we pair each audio utterance with its punctuated transcription from LibriSpeech-PC [44]. In preliminary experiments, we observe that Whisper becomes less stable when transcribing utterances that contain both questions and answer choices. To improve ASR robustness in QA settings, we further augment the training data with speech versions of CommonsenseQA [45], synthesized using Kokoro TTS with the af-heart voice ². We train WhisperPro for 2 epochs with a global batch size of 16, a learning rate of 1×10^{-5} , and a linear decay scheduler with 500 warmup steps.

²<https://github.com/hexgrad/kokoro-82M>

3.2 Prosody-Aware LLM

Our LLM backbone is designed to simultaneously minimize the semantic modality gap and preserve strong paralinguistic understanding. We build on instruction-tuned TLMs, specifically Qwen2.5-3B-Instruct and Qwen2.5-7B-Instruct at different model scales.

Input Format. Our key design principle is to convert speech into a representation that better matches the native input format of a text LLM, while keeping semantic information dominant. Given the transcript and prosody embeddings produced by WhisperPro, we feed the transcript directly into the LLM as the primary input and inject a compressed prosody representation alongside it. We consider two input organization strategies: **1) Global prepending (default):** We summarize prosody into a single utterance-level vector and prepend it to the textual input. As we show later, this compact representation is already sufficient for strong paralinguistic understanding. **2) Interleaving:** To account for cases where prosodic changes within an utterance may matter, we additionally provide an interleaved formulation that inserts compressed prosody embeddings among text tokens at a fixed ratio. These designs preserve the original capabilities of the LLM while still providing a compact representation of how the utterance was spoken, consistent with prior findings that semantic information plays a more central role than raw acoustic detail in SLM modeling [39, 40].

Let the WhisperPro produce a transcription token sequence $\mathbf{y}^{\text{enc}} = (y_1^{\text{enc}}, \dots, y_N^{\text{enc}})$ and the aligned prosody embeddings $\mathbf{P} = (p_1, \dots, p_N)$. We then tokenize the transcribed text again using the LLM tokenizer, yielding $\mathbf{y}^{\text{llm}} = (y_1^{\text{llm}}, \dots, y_T^{\text{llm}})$, where in general $T \neq N$ because the tokenization schemes of WhisperPro and the LLM need not coincide. We instantiate two ways of injecting prosody into the LLM, corresponding to the global prepending and interleaving designs introduced above. For **global prepending**, we first aggregate all prosody embeddings from the user question into a single utterance-level vector by average pooling:

$$\bar{\mathbf{p}} = \frac{1}{N} \sum_{i=1}^N p_i \in \mathbb{R}^d. \quad (7)$$

We then project the averaged prosody embedding into the LLM embedding space using a two-layer Multi-Layer Perceptron (MLP), and place this projected prosody embedding at the front of the input sequence. Therefore, the final input to the LLM is

$$\mathbf{H}_{\text{global}}^{(0)} = [\tilde{\mathbf{p}}; y_1^{\text{llm}}; \dots; y_T^{\text{llm}}], \quad (8)$$

where $\tilde{\mathbf{p}}$ denotes the projected global prosody embedding. Thus, the backbone receives a sequence that is almost identical to a standard text-only prompt, except for one prepended prosody token summarizing how the question was spoken. For **interleaving**, we preserve finer-grained prosodic variation by first determining how many prosody embeddings will be inserted into the LLM input, and then compressing the original prosody sequence into that many averaged embeddings. Given a fixed insertion ratio r , we calculate $M = \lceil T/r \rceil$ as the number of interleaved groups needed for the LLM token sequence, and then evenly partition the original prosody sequence $\mathbf{P}$ of length N into M consecutive groups. For each group, we average the prosody embeddings within that group and apply the same projector to obtain one interleaved prosody embedding $\tilde{p}_j$, resulting in $\tilde{\mathbf{P}} = (\tilde{p}_1, \dots, \tilde{p}_M)$. Finally, the projected prosody embeddings and text tokens are interleaved, giving the final input sequence represented as

$$\mathbf{H}_{\text{interleave}}^{(0)} = [\tilde{p}_1; \mathbf{y}^{(1)}; \tilde{p}_2; \mathbf{y}^{(2)}; \dots; \tilde{p}_M; \mathbf{y}^{(M)}], \quad (9)$$

where $\mathbf{y}^{(j)}$ denotes the j -th LLM text-token group, containing up to r tokens.

Table 1: Summary of the datasets used in WhisperPro and LLM training, where Recon, KD, SER, and AGE represent Reconstruction, Knowledge Distillation, Speech Emotion Recognition, and Age Detection, respectively.

Datasets	Objective	Duration (h)
WhisperPro		
CommonsenseQA [45]	ASR + Recon	40
LibriSpeech [43]	ASR + Recon	960
Total	-	1,000
LLM		
CommonsenseQA [45]	KD	40
UltraChat (part) [46]	KD	255
ParaSpeechCaps [47]	Captioning	342
IEMOCAP [48]	SER	12
CREMA-D [49]	SER	5
SAVEE [50]	SER	0.5
TESS [51]	SER	1.6
ESD (Eng) [52]	SER	14.5
Common Voice (part) [53]	AGE	319
Total	-	989.6

Table 2: ASR and paralinguistic probing performance of WhisperPro compared to Whisper-large-v3. Probing results are reported for 1-layer and 2-layer MLP probes.

Model	Probe	WER ↓		SER	Probing ↑			Avg.
		LibriSpeech	GigaSpeech		SI	F0	Energy	
Whisper-large-v3	1-layer	4.43%	13.66%	33.84	38.83	27.45	14.64	28.69
	2-layer			41.02	60.67	34.09	18.35	38.53
WhisperPro	1-layer	1.96%	12.89%	53.52	79.31	46.81	30.78	52.61
	2-layer			69.63	87.84	50.34	36.56	61.09

Remark 1 We omit the training of the speech synthesis module in this work due to the following two reasons. 1) Dominant approaches in this field measure the modality gap using textual outputs. 2) We see speech synthesis based on textual outputs as a largely decoupled problem whose primary goal is to improve speech naturalness and fidelity, rather than to reduce the modality gap between speech understanding and text understanding.

Training Data and Pipeline. We utilize two sets of training data for different purposes.

1. **Semantic Intelligence Preservation:** Following prior work [11, 31], we construct training data through knowledge distillation from teacher models. Specifically, we use UltraChat³ [46] to distill general instruction-following ability and CommonsenseQA [45] to distill QA ability. We use the original TLMs as teacher models, and the distillation pipeline is as follows. First, we use Kokoro TTS to convert the text-only datasets into speech. Next, we pass the synthesized speech through WhisperPro to obtain transcriptions and prosody embeddings. We then use the transcriptions to query the teacher models and collect their responses. Finally, we feed the transcription-prosody pairs into the LLM backbone as inputs and use the teacher responses as supervision targets.
2. **Paralinguistic Understanding Capabilities:** To develop paralinguistic understanding, we focus on four representative speech attributes: speaker emotion, gender, accent, and age. We use two types of datasets for paralinguistic training. 1) **Speech Captioning:** Speech captioning datasets pair speech utterances with natural-language descriptions that often capture paralinguistic cues. Training on such data helps the model learn to associate speech signals with the paralinguistic information they convey. We use the ParaSpeechCaps dataset [47] for this purpose since it contains the gender and accent attributes. 2) **Feature Detection:** Feature detection datasets provide speech utterances together with ground-truth annotations for a target paralinguistic attribute. Because the captions in ParaSpeechCaps do not include emotion or age labels, we use dedicated feature detection datasets to train the model on these two attributes. For emotion, we use IEMOCAP [48], CREMA-D [49], SAVEE [50], TESS [51], and the English portion of ESD [52]. For age, we use the first 1/4 of the Common Voice data [53] with speaker-age labels.

Remark 2 We acknowledge that, although the features described above are representative, they do not cover the full paralinguistic space. Nevertheless, we use them as a proof of concept to show that our model can demonstrate strong paralinguistic understanding capabilities.

We adopt a two-stage training pipeline. In the first stage, only the prosody projector is trained while the LLM backbone remains frozen. In the second stage, we jointly train the projector and the LLM. Each stage trains for 2 epochs with a global batch size of 1,024 and a linear decay scheduler with no warmups. We use a learning rate of 1×10^{-4} in Stage 1 and 1×10^{-5} in Stage 2. Both stages are trained with cross-entropy loss. Table 1 summarizes the datasets used to train WhisperPro and the LLM, along with their training objectives and audio durations.

4 Experiments

We conduct experiments to demonstrate that our proposed approach minimizes the modality gap while demonstrating strong paralinguistic understanding abilities. We provide both WhisperPro and LLM evaluation results.

³We use only 1/10 of UltraChat, i.e., train_2.jsonl, from <https://huggingface.co/datasets/openbmb/ultrachat>.

Table 3: Various SLM performance $\uparrow$ (%) and modality gap $\downarrow$ on different benchmarks. Gap cells are color-coded from best (green) to worst (red) for each task, where lower is better. We save the word Instruct for Kimi-Audio, Qwen2.5-3B and 7B to save spaces.

	StoryCloze		MMSU		OBQA		ARC-Easy		ARC-C		PIQA		Avg.	
	Acc.	Gap	Acc.	Gap	Acc.	Gap	Acc.	Gap	Acc.	Gap	Acc.	Gap	Acc.	Gap
Random	50.0	-	25.0	-	25.0	-	25.0	-	25.0	-	50.0	-	33.3	-
<i>Cascaded Toplines: ASR (Whisper-v3-Large) + LLM</i>														
ASR + Qwen2.5-3B	88.3	4.3	58.9	2.7	72.7	4.0	90.5	3.7	80.1	3.7	52.7	-0.6	73.9	3.0
ASR + Qwen2.5-7B	89.1	0.1	67.9	2.9	80.0	5.7	89.9	6.2	83.3	5.8	63.0	5.1	78.9	4.3
<i>End-to-end Systems</i>														
Qwen2-Audio-7B	71.9	9.0	29.5	18.7	39.6	37.1	43.5	28.5	43.5	28.5	73.4	5.4	50.2	21.2
DiVA-Llama3.1-8B	82.1	15.8	28.8	34.1	40.0	35.0	39.7	53.6	33.7	48.9	35.6	30.9	43.3	36.4
GLM-4-Voice-9B	76.4	20.6	39.2	28.3	52.1	32.5	73.2	24.6	59.5	33.8	47.3	30.9	57.9	28.4
Qwen2.5-Omni-7B	83.9	5.4	61.3	9.6	81.5	4.2	94.9	1.2	86.1	3.0	72.8	-4.8	80.1	3.1
Kimi-Audio-7B	66.6	22.6	61.6	9.2	83.7	2.0	95.2	0.9	88.2	1.0	82.9	-14.8	79.7	3.5
<i>Modality Gap-optimized Systems</i>														
SALAD-3B														
Stage I	75.5	7.4	47.3	14.6	65.5	16.3	-	-	75.6	6.2	78.3	0.3	68.4	9.0
Stage II	75.8	7.1	52.5	9.4	76.7	5.1	-	-	79.9	1.9	78.1	0.5	72.6	4.8
TextPro-SLM-3B	84.2	8.4	57.6	4.0	73.4	3.3	92.9	1.4	80.7	3.1	56.9	-4.8	74.3	2.6
SALAD-7B														
Stage I	81.5	3.5	55.3	15.5	69.7	19.3	-	-	82.3	6.1	80.3	0.4	73.8	9.0
Stage II	81.5	3.5	57.5	13.3	75.1	13.9	-	-	84.0	4.4	80.3	0.4	75.7	7.1
TextPro-SLM-7B	88.6	0.6	66.7	4.2	85.1	0.7	95.8	0.3	89.4	-0.2	69.7	-1.6	82.5	0.7
TextPro-SLM-7B 5:1	87.6	1.6	66.2	4.7	83.3	2.5	95.9	0.2	88.8	0.4	70.8	-2.7	82.1	1.1
<i>Ablation Studies</i>														
TextPro-SLM-7B w/o KD	84.7	4.5	46.7	24.2	55.8	30.0	89.5	6.6	75.8	13.4	66.6	1.5	69.9	13.3
WhisperPro + Qwen2.5-7B	88.8	0.4	62.6	8.3	80.4	5.4	95.0	1.1	87.8	1.4	63.3	4.8	79.7	3.6

4.1 WhisperPro Evaluations

Before evaluating the full system, we first assess WhisperPro in isolation. Our goal is to verify that it preserves Whisper’s strong semantic modeling ability while learning richer prosodic representations. To this end, we compare WhisperPro directly with the original Whisper-large-v3 along two dimensions: ASR quality and paralinguistic information encoded in the prosody embeddings.

To evaluate semantic modeling, we measure ASR performance on the test sets of LibriSpeech [43] and GigaSpeech [54] using word error rate (WER). As shown in Table 2, WhisperPro achieves slightly lower WER than Whisper-large-v3 on both benchmarks. This result indicates that our modifications do not compromise the strong transcription ability of the original Whisper model.

To evaluate prosodic modeling, we conduct probing experiments on four paralinguistic attributes: speech emotion recognition (SER), speaker identification (SI), fundamental frequency (F0), and energy. We use RAVDESS [55] for SER, the VoxCeleb1 test set [56] for SI, and the LibriTTS test set [57] for F0 and energy. Following standard probing practice, we train both 1-layer and 2-layer MLP classifiers on top of the extracted prosody embeddings. Because F0 and energy are continuous-valued signals rather than categorical labels, we discretize each feature into 10 quantile-based bins. More experimental details of the probing experiments are presented in Appendix C. Table 2 shows that the prosody embeddings produced by WhisperPro consistently encode substantially richer information than the original Whisper representations across all four attributes. Together, these findings suggest that our reconstruction objective significantly enhances paralinguistic representation learning while preserving semantic fidelity.

4.2 LLM Evaluations

4.2.1 Experimental Setup

Benchmarks and Evaluations. We evaluate our LLM on two groups of benchmarks: one for measuring the modality gap and the other for assessing paralinguistic understanding. To evaluate the modality gap, we follow the benchmark setup of prior work [11], including Spoken StoryCloze [14],

OpenBookQA and MMSU from VoiceBench [58], PIQA [59], and ARC, including both the Easy and Challenge splits [60]. To evaluate paralinguistic understanding, we primarily use the paralinguistic task set from AIR-Bench [61], which covers emotion recognition, speaker gender recognition, and speaker age prediction. All evaluations are conducted in the zero-shot setting.

In addition to the paralinguistic tasks provided by AIR-Bench, we consider speaker accent as another important paralinguistic attribute. Since we did not find a convenient off-the-shelf benchmark for this task, we construct our own accent detection benchmark and evaluate our models on it. Specifically, we use a subset of the Common Voice test set with accent labels and convert it into an AIR-Bench-style four-option multiple-choice benchmark. The four target labels are “United States”, “England”, “India and South Asia (India, Pakistan, Sri Lanka)”, and “Europe”. We sample 1,000 examples to build the benchmark. Details of the accent benchmark construction are provided in Appendix D.⁴

Baselines. We compare our model with a diverse set of SLM baselines, including leading industrial systems and methods explicitly designed to reduce the modality gap. We calculate the modality gap as the performance difference between a speech model given spoken input and its backbone TLM given textual input. The compared model pairs are GLM-4-Voice [12] and GLM-4-9B [10], Qwen2-Audio [62] and Qwen-7B-Chat [63], DiVA [64] and Llama-3-8B [65], Qwen2.5-Omni [23] and Qwen2.5-7B-Instruct [66], Kimi-Audio [29] and Qwen2.5-7B-Instruct, and SALAD [11] with Qwen2.5-3B&7B-Base. As a reference point, we also include cascaded systems that combine Whisper-large-v3 with Qwen2.5-3B&7B-Instruct.

4.2.2 Modality Gap Results

Following Cuervo *et al.* [11], we report both accuracy and modality gap for each dataset and model. The results are summarized in Table 3, where TextPro-SLM denotes the global prepending setting and TextPro-SLM 5:1 denotes the interleaving setting with a 5:1 ratio. We make three main observations. **1) TextPro-SLM achieves the lowest average modality gap among all compared SLMs at both the 3B and 7B scales.** In particular, TextPro-SLM-7B attains an average modality gap of 0.7, substantially lower than SALAD’s 7.1 and Qwen2.5-Omni’s 3.1. **2) TextPro-SLM-7B 5:1 achieves performance comparable to that of TextPro-SLM-7B.** This result suggests that, even with more injected prosody embeddings, our method preserves semantic capabilities effectively. **3) TextPro-SLM consistently achieves a slightly lower modality gap than the cascaded topline.** On the 7B setting, it yields a lower modality gap on 4 of 6 benchmarks. We suspect that this advantage comes from the dedicated knowledge distillation applied to WhisperPro’s outputs, and we examine this hypothesis further in the ablation study. **4) TextPro-SLM shows particularly strong gains on reasoning-required benchmarks compared with SALAD.** Datasets such as MMSU and OBQA require stronger reasoning ability, and TextPro-SLM-7B reduces the gap to 4.2 and 0.7, respectively, compared with SALAD’s 13.3 and 13.9. Overall, these observations suggest that reducing the modality gap from the input side is highly effective, particularly considering TextPro-SLM only requires **fewer than 300 hours of audio** for knowledge distillation.

Ablation Studies. We study two ablated variants in Table 3. **1) TextPro-SLM-7B w/o KD:** In this setting, we remove knowledge distillation and retain only the paralinguistic-task training. As expected, performance drops substantially. **2) WhisperPro + Qwen2.5-7B:** In this setting, we directly feed the text tokens produced by WhisperPro to Qwen2.5-7B-Instruct. We find that performance remains broadly comparable to that of TextPro-SLM, with slight drops on some benchmarks. Because the main differences between this setting and TextPro-SLM are the insertion of prosody embeddings and end-to-end model training, we hypothesize that TextPro-SLM benefits from adapting the LLM to WhisperPro-produced texts, making the backbone more compatible with the speech encoder.

Table 4: Reasoning-heavy VoxEval math performance $\uparrow$ and modality gap $\downarrow$ (%).

	Elementary		High School		College		Avg.	
	Acc.	Gap	Acc.	Gap	Acc.	Gap	Acc.	Gap
<i>Text Topline</i>								
Qwen2.5-7B-Instruct	83.2	-	73.1	-	63.0	-	73.1	-
<i>End-to-end Systems</i>								
Qwen2.5-Omni-7B	74.3	8.9	62.6	10.5	47.8	15.2	61.6	11.5
Kimi-Audio-7B	66.1	17.1	55.6	17.5	26.1	37.0	49.2	23.9
TextPro-SLM-7B	80.0	3.2	71.3	1.8	52.2	10.8	67.8	5.3

⁴We will open-source this accent benchmark.

Modality Gap on Reasoning-Heavy Tasks. Although our benchmark choices for measuring the modality gap in Table 3 follow prior work, they mostly involve tasks with low to moderate reasoning demands. We therefore further examine the modality gap on more reasoning-heavy benchmarks. Specifically, we use the three math-related tasks in VoxEval [67]: elementary math, high school math, and college math. Because Qwen2.5-Omni-7B and Kimi-Audio-7B show the lowest modality gap among the baseline SLMs in Table 3, we compare these two models with our approach on these three benchmarks. We cap the computational budget at 512 tokens. The results in Table 4 highlight two main findings. **1) All SLMs exhibit a larger modality gap on reasoning-heavy tasks.** Specifically, the average modality gap of Qwen2.5-Omni-7B increases from 3.1% to 11.5%, while Kimi-Audio-7B shows an average gap above 20%. **2) Our approach performs substantially better on reasoning-heavy tasks.** In particular, TextPro-SLM achieves a modality gap of just 1.8% on high school math, compared with 10.5% for Qwen2.5-Omni-7B and 17.5% for Kimi-Audio-7B. Overall, TextPro-SLM also yields the lowest average modality gap across these benchmarks.

4.2.3 Paralinguistic Understanding Results

Table 5 reports the results on four paralinguistic understanding tasks and reveals two main observations. **1) TextPro-SLM achieves the strongest paralinguistic understanding performance among all baselines.** This result suggests that even with only a single prosody embedding and roughly 1,000 hours of training audio, an SLM can already achieve state-of-the-art paralinguistic understanding. **2) TextPro-SLM-5:1 performs slightly better than TextPro-SLM on paralinguistic understanding.** In particular, it improves performance on three of the four tasks, likely because feeding more prosody embeddings into the LLM exposes richer acoustic cues. However, the overall gain remains modest, indicating that simply increasing the amount of prosodic input does not substantially improve paralinguistic understanding. Taken together, these results support our hypothesis that, for SLMs, semantic information is more critical than additional acoustic detail.

Ablation Studies. We study two ablated variants in Table 5. **1) Without the reconstruction objective in WhisperPro (TextPro-SLM-7B w/o Recon):** In this setting, we use the prosody embedding from the original Whisper-large-v3, without our reconstruction-based training. Performance drops on all four paralinguistic tasks, indicating that the reconstruction objective is critical for preserving paralinguistic information throughout the pipeline. **2) Without paralinguistic training (TextPro-SLM-7B w/o Train):** In this setting, we directly feed the prosody embedding from WhisperPro into Qwen2.5-7B-Instruct without training the model on paralinguistic tasks. This variant also shows a substantial performance drop, confirming TextPro-SLM effectively acquires paralinguistic information through task-specific training.

Table 5: SLM performance (%) on various paralinguistic understanding tasks.

Model	Emo	Gender	Age	Accent	AVG
Qwen2-Audio-7B	48.2	64.7	23.1	29.0	41.3
DiVA	30.7	46.8	33.3	32.5	35.8
GLM-4-Voice-9B	23.0	23.9	18.7	32.4	24.5
Qwen2.5-Omni-7B	54.8	89.8	44.8	45.3	58.7
Kimi-Audio-7B	61.1	75.9	60.5	31.2	57.2
TextPro-SLM-3B	58.8	80.2	63.7	45.0	61.9
TextPro-SLM-7B	60.5	88.6	64.9	45.1	64.8
TextPro-SLM-7B 5:1	57.7	88.8	66.7	50.2	65.8
<i>Ablation Studies</i>					
TextPro-SLM-7B w/o Recon	56.7	79.9	64.2	44.4	61.3
TextPro-SLM-7B w/o Train	51.8	28.5	43.6	16.1	35.0

5 Conclusion

In this work, we propose TextPro-SLM, a speech LLM designed to minimize the modality gap from the input side by making spoken input resemble that of a prosody-aware text LLM. Our approach combines WhisperPro, which produces synchronized text tokens and prosody embeddings, with an LLM backbone that preserves the semantic strengths of the original text model while learning to interpret paralinguistic information. Across experiments at both 3B and 7B scales, TextPro-SLM achieves the lowest modality gap among strong SLM baselines and delivers strong performance on paralinguistic understanding benchmarks, while requiring only about 1,000 hours of audio training data. These results show that aligning speech inputs with text-style representations, with prosodic information preserved, is a promising and data-efficient direction for building more capable and practical SLMs.

References

- [1] Wenqian Cui, Dianzhi Yu, Xiaoqi Jiao, Ziqiao Meng, Guangyan Zhang, Qichao Wang, Steven Y Guo, and Irwin King. Recent advances in speech language models: A survey. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13943–13970, 2025.
- [2] Qingkai Fang, Shoutao Guo, Yan Zhou, Zhengrui Ma, Shaolei Zhang, and Yang Feng. Llama-omni: Seamless speech interaction with large language models. *arXiv preprint arXiv:2409.06666*, 2024.
- [3] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.
- [4] Wenqian Cui, Lei Zhu, Xiaohui Li, Zhihan Guo, Haoli Bai, Lu Hou, and Irwin King. Think before you talk: Enhancing meaningful dialogue generation in full-duplex speech language models with planning-inspired text guidance. *arXiv preprint arXiv:2508.07375*, 2025.
- [5] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [6] Aaditya Singh, Adam Fry, Adam Perelman, Adam Tart, Adi Ganesh, Ahmed El-Kishky, Aidan McLaughlin, Aiden Low, AJ Ostrow, Akhila Ananthram, et al. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267*, 2025.
- [7] Aaron Adcock, Aayushi Srivastava, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pande, Abhinav Pandey, Abhinav Sharma, Abhishek Kadian, Abhishek Kumawat, Adam Kelsey, et al. The llama 4 herd: Architecture, training, evaluation, and deployment notes. *arXiv preprint arXiv:2601.11659*, 2026.
- [8] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- [9] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- [10] Team Glm, Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Dan Zhang, Diego Rojas, Guanyu Feng, Hanlin Zhao, et al. Chatglm: A family of large language models from glm-130b to glm-4 all tools. *arXiv preprint arXiv:2406.12793*, 2024.
- [11] Santiago Cuervo, Skyler Seto, Maureen de Seyssel, Richard He Bai, Zijin Gu, Tatiana Likhomanenko, Navdeep Jaitly, and Zakaria Aldeneh. Closing the gap between text and speech understanding in llms. *arXiv preprint arXiv:2510.13632*, 2025.
- [12] Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong Wang, Shengmin Jiang, Lei Zhao, Yuxiao Dong, and Jie Tang. Glm-4-voice: Towards intelligent and human-like end-to-end spoken chatbot. *arXiv preprint arXiv:2412.02612*, 2024.
- [13] Kushal Lakhotia, Eugene Kharonov, Wei-Ning Hsu, Yossi Adi, Adam Polyak, Benjamin Bolte, Tu-Anh Nguyen, Jade Copet, Alexei Baevski, Abdelrahman Mohamed, et al. On generative spoken language modeling from raw audio. *Transactions of the Association for Computational Linguistics*, 9:1336–1354, 2021.
- [14] Michael Hassid, Tal Remez, Tu Anh Nguyen, Itai Gat, Alexis Conneau, Felix Kreuk, Jade Copet, Alexandre Defossez, Gabriel Synnaeve, Emmanuel Dupoux, et al. Textually pretrained speech language models. *Advances in Neural Information Processing Systems*, 36, 2024.
- [15] Tu Anh Nguyen, Benjamin Muller, Bokai Yu, Marta R Costa-Jussa, Maha Elbayad, Sravya Popuri, Paul-Ambroise Duquenne, Robin Algayres, Ruslan Mavlyutov, Itai Gat, et al. Spirit-lm: Interleaved spoken and written language model. *arXiv preprint arXiv:2402.05755*, 2024.

- [16] Dong Zhang, Shimin Li, Xin Zhang, Jun Zhan, Pengyu Wang, Yaqian Zhou, and Xipeng Qiu. SpeechGPT: Empowering large language models with intrinsic cross-modal conversational abilities. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 15757–15773, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.1055. URL <https://aclanthology.org/2023.findings-emnlp.1055>.
- [17] Zhifei Xie and Changqiao Wu. Mini-omni: Language models can hear, talk while thinking in streaming. *arXiv preprint arXiv:2408.16725*, 2024.
- [18] Zuwei Long, Yunhang Shen, Chaoyou Fu, Heting Gao, Peixian Chen, Mengdan Zhang, Hang Shao, Jian Li, Jinlong Peng, Haoyu Cao, et al. Vita-audio: Fast interleaved audio-text token generation for efficient large speech-language model. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [19] Qingkai Fang, Yan Zhou, Shoutao Guo, Shaolei Zhang, and Yang Feng. Llama-omni 2: Llm-based real-time spoken chatbot with autoregressive streaming speech synthesis. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 18617–18629, 2025.
- [20] Qian Chen, Yafeng Chen, Yanni Chen, Mengzhe Chen, Yingda Chen, Chong Deng, Zhihao Du, Ruize Gao, Changfeng Gao, Zhifu Gao, et al. Minmo: A multimodal large language model for seamless voice interaction. *arXiv preprint arXiv:2501.06282*, 2025.
- [21] Yuhao Wang, Heyang Liu, Ziyang Cheng, Ronghua Wu, Qunshan Gu, Yanfeng Wang, and Yu Wang. Vocalnet: Speech llms with multi-token prediction for faster and high-quality generation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 19595–19612, 2025.
- [22] Jin Xu, Zhifang Guo, Hangrui Hu, Yunfei Chu, Xiong Wang, Jinzheng He, Yuxuan Wang, Xian Shi, Ting He, Xinfu Zhu, et al. Qwen3-omni technical report. *arXiv preprint arXiv:2509.17765*, 2025.
- [23] Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and Junyang Lin. Qwen2.5-omni technical report, 2025. URL <https://arxiv.org/abs/2503.20215>.
- [24] Bajian Xiang, Shuaijiang Zhao, Tingwei Guo, and Wei Zou. Understanding the modality gap: An empirical study on the speech-text alignment mechanism of large speech language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 5187–5202, 2025.
- [25] Yuhao Zhang, Zhiheng Liu, Fan Bu, Ruiyu Zhang, Benyou Wang, and Haizhou Li. Soundwave: Less is more for speech-text alignment in llms. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 18718–18738, 2025.
- [26] Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in neural information processing systems*, 33:12449–12460, 2020.
- [27] Yu-An Chung, Yu Zhang, Wei Han, Chung-Cheng Chiu, James Qin, Ruoming Pang, and Yonghui Wu. W2v-bert: Combining contrastive learning and masked language modeling for self-supervised speech pre-training. In *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 244–250. IEEE, 2021.
- [28] Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. Hubert: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM transactions on audio, speech, and language processing*, 29:3451–3460, 2021.
- [29] Ding Ding, Zeqian Ju, Yichong Leng, Songxiang Liu, Tong Liu, Zeyu Shang, Kai Shen, Wei Song, Xu Tan, Heyi Tang, et al. Kimi-audio technical report. *arXiv preprint arXiv:2504.18425*, 2025.

- [30] Ming-Hao Hsu, Xueyao Zhang, Xiaohai Tian, Jun Zhang, and Zhizheng Wu. Anatomy of the modality gap: Dissecting the internal states of end-to-end speech llms. *arXiv preprint arXiv:2603.01502*, 2026.
- [31] Enzhi Wang, Qicheng Li, Zhiyuan Tang, and Yuhang Jia. Cross-modal knowledge distillation for speech large language models. *arXiv preprint arXiv:2509.14930*, 2025.
- [32] Di Cao, Dongjie Fu, Hai Yu, Siqi Zheng, Xu Tan, and Tao Jin. X-opd: Cross-modal on-policy distillation for capability alignment in speech llms. *arXiv preprint arXiv:2603.24596*, 2026.
- [33] Jing Hu, Danxiang Zhu, Xianlong Luo, Dan Zhang, Shuwei He, Yishu Lei, Haitao Zheng, Shikun Feng, Jingzhou He, Yu Sun, et al. Cord: Bridging the audio-text reasoning gap via weighted on-policy cross-modal distillation. *arXiv preprint arXiv:2601.16547*, 2026.
- [34] Runyan Yang, Yuke Si, Yingying Gao, Junlan Feng, Chao Deng, and Shilei Zhang. Teaching audio models to reason: A unified framework for source-and layer-wise distillation. *arXiv preprint arXiv:2509.18579*, 2025.
- [35] Hang Shao, Heting Gao, Yunhang Shen, Jiawei Chen, Zuwei Long, Dong Yang, Ke Li, and Xing Sun. Deepomni: Towards seamless and smart speech interaction with adaptive modality-specific moe. *arXiv preprint arXiv:2506.21864*, 2025.
- [36] Siyuan Mu and Sen Lin. A comprehensive survey of mixture-of-experts: Algorithms, theory, and applications. *arXiv preprint arXiv:2503.07137*, 2025.
- [37] Chi-Yuan Hsiao, Ke-Han Lu, Kai-Wei Chang, Chih-Kai Yang, Wei-Chih Chen, and Hung-yi Lee. Analyzing mitigation strategies for catastrophic forgetting in end-to-end training of spoken language models. *arXiv preprint arXiv:2505.17496*, 2025.
- [38] Dianzhi Yu, Xinni Zhang, Yankai Chen, Aiwei Liu, Yifei Zhang, Philip S Yu, and Irwin King. Recent advances of multimodal continual learning: A comprehensive survey. *arXiv preprint arXiv:2410.05352*, 2024.
- [39] Zalán Borsos, Raphaël Marinier, Damien Vincent, Eugene Kharitonov, Olivier Pietquin, Matt Sharifi, Dominik Roblek, Olivier Teboul, David Grangier, Marco Tagliasacchi, et al. Audioldm: a language modeling approach to audio generation. *IEEE/ACM transactions on audio, speech, and language processing*, 31:2523–2533, 2023.
- [40] Amir Hussein, Sameer Khurana, Gordon Wicern, Francois G Germain, and Jonathan Le Roux. Hasrd: Hierarchical acoustic and semantic representation disentanglement. *arXiv preprint arXiv:2506.00843*, 2025.
- [41] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR, 2023.
- [42] Shengpeng Ji, Ziyue Jiang, Wen Wang, Yifu Chen, Minghui Fang, Jialong Zuo, Qian Yang, Xize Cheng, Zehan Wang, Ruiqi Li, et al. Wavtokenizer: an efficient acoustic discrete codec tokenizer for audio language modeling. *arXiv preprint arXiv:2408.16532*, 2024.
- [43] Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. Librispeech: an asr corpus based on public domain audio books. In *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 5206–5210. IEEE, 2015.
- [44] Aleksandr Meister, Matvei Novikov, Nikolay Karpov, Evelina Bakhturina, Vitaly Lavruchin, and Boris Ginsburg. Librispeech-pc: Benchmark for evaluation of punctuation and capitalization capabilities of end-to-end asr models. In *2023 IEEE automatic speech recognition and understanding workshop (ASRU)*, pages 1–7. IEEE, 2023.
- [45] Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. Commonsenseqa: A question answering challenge targeting commonsense knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4149–4158, 2019.

- [46] Ning Ding, Yulin Chen, Bokai Xu, Yujia Qin, Shengding Hu, Zhiyuan Liu, Maosong Sun, and Bowen Zhou. Enhancing chat language models by scaling high-quality instructional conversations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 3029–3051, 2023.
- [47] Anuj Diwan, Zhisheng Zheng, David Harwath, and Eunsol Choi. Scaling rich style-prompted text-to-speech datasets. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 3639–3659, 2025.
- [48] Carlos Busso, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeannette N Chang, Sungbok Lee, and Shrikanth S Narayanan. Iemocap: Interactive emotional dyadic motion capture database. *Language resources and evaluation*, 42(4):335–359, 2008.
- [49] Houwei Cao, David G Cooper, Michael K Keutmann, Ruben C Gur, Ani Nenkova, and Ragini Verma. Crowd-sourced emotional multimodal actors dataset (crema-d), 2025.
- [50] Philip Jackson and SJUoSG Haq. Surrey audio-visual expressed emotion (savee) database. *University of Surrey: Guildford, UK*, 2014.
- [51] Kate Dupuis and M Kathleen Pichora-Fuller. *Toronto emotional speech set (TESS)*. University of Toronto, Psychology Department Toronto, ON, Canada, 2010.
- [52] Kun Zhou, Berrak Sisman, Rui Liu, and Haizhou Li. Emotional voice conversion: Theory, databases and esd. *Speech Communication*, 137:1–18, 2022.
- [53] Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Moraes, Lindsay Saunders, Francis Tyers, and Gregor Weber. Common voice: A massively-multilingual speech corpus. In *Proceedings of the twelfth language resources and evaluation conference*, pages 4218–4222, 2020.
- [54] Guoguo Chen, Shuzhou Chai, Guanbo Wang, Jiayu Du, Wei-Qiang Zhang, Chao Weng, Dan Su, Daniel Povey, Jan Trmal, Junbo Zhang, et al. Gigaspeech: An evolving, multi-domain asr corpus with 10,000 hours of transcribed audio. *arXiv preprint arXiv:2106.06909*, 2021.
- [55] Steven R Livingstone and Frank A Russo. The ryerson audio-visual database of emotional speech and song (ravdess): A dynamic, multimodal set of facial and vocal expressions in north american english. *PloS one*, 13(5):e0196391, 2018.
- [56] Arsha Nagrani, Joon Son Chung, Weidi Xie, and Andrew Zisserman. Voxceleb: Large-scale speaker verification in the wild. *Computer Speech & Language*, 60:101027, 2020.
- [57] Heiga Zen, Viet Dang, Rob Clark, Yu Zhang, Ron J Weiss, Ye Jia, Zhifeng Chen, and Yonghui Wu. Libritts: A corpus derived from librispeech for text-to-speech. *arXiv preprint arXiv:1904.02882*, 2019.
- [58] Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T Tan, and Haizhou Li. Voicebench: Benchmarking llm-based voice assistants. *Transactions of the Association for Computational Linguistics*, 14:378–398, 2026.
- [59] Yonatan Bisk, Rowan Zellers, Jianfeng Gao, Yejin Choi, et al. Piqa: Reasoning about physical commonsense in natural language. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 7432–7439, 2020.
- [60] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.
- [61] Qian Yang, Jin Xu, Wenrui Liu, Yunfei Chu, Ziyue Jiang, Xiaohuan Zhou, Yichong Leng, Yuanjun Lv, Zhou Zhao, Chang Zhou, et al. Air-bench: Benchmarking large audio-language models via generative comprehension. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1979–1998, 2024.
- [62] Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, et al. Qwen2-audio technical report. *arXiv preprint arXiv:2407.10759*, 2024.

- [63] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023.
- [64] William Held, Yanzhe Zhang, Minzhi Li, Weiyang Shi, Michael J Ryan, and Diyi Yang. Distilling an end-to-end voice assistant without instruction training data. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7876–7891, 2025.
- [65] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [66] Qwen Team. Qwen2.5 technical report, 2024. URL <https://arxiv.org/abs/2412.15115>.
- [67] Wenqian Cui, Xiaoqi Jiao, Ziqiao Meng, and Irwin King. Voxeval: Benchmarking the knowledge understanding capabilities of end-to-end spoken language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16735–16753, 2025.
- [68] Eugene Kharitonov, Ann Lee, Adam Polyak, Yossi Adi, Jade Copet, Kushal Lakhotia, Tu-Anh Nguyen, Morgane Riviere, Abdelrahman Mohamed, Emmanuel Dupoux, et al. Text-free prosody-aware generative spoken language modeling. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8666–8681, 2022.
- [69] Dong Zhang, Xin Zhang, Jun Zhan, Shimin Li, Yaqian Zhou, and Xipeng Qiu. Speechgpt-gen: Scaling chain-of-information speech generation. *arXiv preprint arXiv:2401.13527*, 2024.
- [70] Zeqian Ju, Yuanheng Wang, Kai Shen, Xu Tan, Detai Xin, Dongchao Yang, Yanqing Liu, Yichong Leng, Kaitao Song, Siliang Tang, et al. Natralspeech 3: Zero-shot speech synthesis with factorized codec and diffusion models. *arXiv preprint arXiv:2403.03100*, 2024.
- [71] Kaizhi Qian, Xulin Fan, Junrui Ni, Slava Shechtman, Mark Hasegawa-Johnson, Chuang Gan, and Yang Zhang. Prosodylm: Uncovering the emerging prosody processing capabilities in speech language models. *arXiv preprint arXiv:2507.20091*, 2025.

A Discussions & Limitations

Although we have shown promising results for TextPro-SLM, the current study has several limitations. First, we do not train or evaluate a speech synthesis module, and all comparisons are made on text outputs. We make this choice deliberately because modality-gap studies in recent Speech LLM literature typically compare benchmark performance using text outputs, which has become a standard evaluation protocol. Under the increasingly common decoupled “thinking” and “talking” design, we view text generation as the component that most directly reflects the model’s intelligence and understanding ability, while speech synthesis primarily concerns how accurately and naturally the generated text is pronounced. These two aspects are usually measured with different metrics, such as task accuracy for text generation and metrics such as WER and UTMOS for speech synthesis.

Second, although representative, our paralinguistic evaluation does not reflect the whole paralinguistic space. We choose a set of tasks that covers several important paralinguistic phenomena and use them to test whether the model can exploit prosodic information beyond plain lexical content. However, real-world spoken interaction may require understanding more complex and dynamic signals, such as abrupt emotion shifts, mixed affective states, speaker intention changes, or other nuanced discourse-level cues. Therefore, our paralinguistic results should be interpreted as a proof of concept that TextPro-SLM can achieve strong paralinguistic understanding rather than achieving state-of-the-art performance on the whole paralinguistic space.

Third, our current implementation may introduce additional latency. In our setup, the speech encoder runs only after the user has finished speaking, because the Whisper backbone we use does not support streaming inference. As a result, the system is sequential rather than fully streamable, which can slightly increase response latency in interactive scenarios. We view this issue primarily as a limitation

of the underlying Whisper backbone rather than of the core idea of our method. In principle, the same input-side modality-gap reduction strategy could be combined with a streaming ASR backbone to support lower-latency processing. We leave this direction to future work.

Finally, adapting our current framework to non-speech audio is non-trivial. Our current pipeline explicitly decomposes the input into a text transcription stream and a prosody embedding stream, which is well-suited to spoken language but not to general audio events. In more complex settings, a model may need to interpret non-speech sounds or mixed audio inputs that cannot be naturally represented through standard transcription. Extending the framework to such cases will likely require dedicated training data and new modeling techniques. One possible direction is to let the text stream emit placeholder tokens, such as masked tokens, when the input contains non-speech audio, while relying on the prosody or acoustic embedding stream to preserve the relevant information. We consider this extension important but outside the scope of the present paper.

B Detailed Architecture Explanation of WhisperPro

Input and Whisper backbone. The architecture of WhisperPro includes two main components: Whisper-large-v3 (encoder) and Mel Reconstructor (decoder). Let an input utterance be a raw waveform $\mathbf{x} \in \mathbb{R}^T$, where T denotes the number of waveform samples. Following Whisper’s standard preprocessing pipeline, the waveform is first resampled and normalized to a fixed 30-second window, and then converted into a log-Mel spectrogram $\mathbf{M} \in \mathbb{R}^{F \times L}$, where $F = 128$ is the number of Mel bins and $L = 3000$ is the number of time frames. Thus, regardless of the original utterance duration, the acoustic input to WhisperPro is always represented as a 128×3000 Mel spectrogram.

Whisper-large-v3 then processes $\mathbf{M}$ using its encoder-decoder Transformer architecture. First, the Whisper encoder maps the acoustic input into a sequence of latent speech representations,

$$\mathbf{H}^{\text{enc}} = \text{Enc}_{\text{whisper}}(\mathbf{M}) \in \mathbb{R}^{L_e \times d}, \quad (10)$$

where L_e denotes the encoder sequence length and d is the Whisper hidden dimension. Conditioned on $\mathbf{H}^{\text{enc}}$, the Whisper decoder autoregressively predicts a text token sequence

$$\mathbf{y} = (y_1, \dots, y_N), \quad y_i \in \mathcal{V}, \quad (11)$$

with $\mathcal{V}$ the text vocabulary and N the output token length. Concretely, at decoder layer l , the hidden states are represented as

$$\mathbf{H}^{(l)} = (\mathbf{h}_1^{(l)}, \dots, \mathbf{h}_N^{(l)}) \in \mathbb{R}^{N \times d}. \quad (12)$$

In our model, the prosody embeddings are taken from the 16-th Whisper decoder layer:

$$\mathbf{P} = \mathbf{H}^{(16)} \in \mathbb{R}^{N \times d}. \quad (13)$$

Because $\mathbf{P}$ is extracted from the decoder states aligned with the generated text sequence, each prosody vector $\mathbf{p}_i$ is naturally paired with exactly one text token y_i . Therefore, the model represents spoken input as two synchronized streams:

$$\{(y_i, \mathbf{p}_i)\}_{i=1}^N, \quad (14)$$

where y_i captures the linguistic content and $\mathbf{p}_i$ captures the associated prosodic information.

Mel reconstructor. The Mel Reconstructor is designed to recover the original Mel spectrogram $\mathbf{M}$ from these two token-aligned streams. Its role is to force the decoder hidden states to preserve information about how the utterance was spoken, rather than only what was said. To this end, each text token y_i is first embedded into a continuous vector,

$$\mathbf{e}_i = \text{Emb}(y_i) \in \mathbb{R}^{d_r}, \quad (15)$$

where d_r is the hidden size of the reconstructor. In parallel, each prosody embedding $\mathbf{p}_i \in \mathbb{R}^d$ is projected into the same space:

$$\tilde{\mathbf{p}}_i = \mathbf{W}_p \mathbf{p}_i + \mathbf{b}_p \in \mathbb{R}^{d_r}. \quad (16)$$

The two representations are then fused token-wise. Specifically, they are concatenated first and then projected by a feed-forward layer:

$$\mathbf{z}_i = \phi([\mathbf{e}_i; \tilde{\mathbf{p}}_i]) \in \mathbb{R}^{d_r}, \quad (17)$$

where $[\cdot; \cdot]$ denotes concatenation and $\phi(\cdot)$ denotes a fusion block consisting of a linear layer, layer normalization, GELU activation, and dropout. Stacking the resulting token-level fused vectors yields

$$\mathbf{Z} = (\mathbf{z}_1, \dots, \mathbf{z}_N) \in \mathbb{R}^{N \times d_r}. \quad (18)$$

To encode contextual dependencies across the whole utterance, $\mathbf{Z}$ is augmented with sinusoidal positional encoding and passed through a Transformer encoder:

$$\mathbf{U} = \text{Enc}_{\text{mel}}(\mathbf{Z}) \in \mathbb{R}^{N \times d_r}. \quad (19)$$

This encoded sequence $\mathbf{U}$ serves as the memory for Mel reconstruction. Intuitively, $\mathbf{U}$ integrates token identity and prosodic realization into a contextualized latent sequence.

The decoder side of the Mel Reconstructor is implemented as a Transformer decoder with cross-attention and a fixed set of learnable frame queries, one for each target Mel frame. This design addresses the mismatch between the variable-length token-level input sequence and the fixed-length frame-level output sequence produced by Whisper’s preprocessing. Rather than imposing a one-to-one alignment between tokens and Mel frames, cross-attention allows each frame query to attend over the full token-level memory and extract the information needed to reconstruct its corresponding acoustic frame. Specifically, we introduce

$$\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_L) \in \mathbb{R}^{L \times d_r}, \quad (20)$$

where $L = 3000$ matches the number of Mel frames in Whisper’s normalized input. We first add the same sinusoidal positional encoding to $\mathbf{Q}$ and feed it into a Transformer decoder that performs cross-attention over the memory $\mathbf{U}$:

$$\mathbf{D} = \text{Dec}_{\text{mel}}(\mathbf{Q}, \mathbf{U}) \in \mathbb{R}^{L \times d_r}. \quad (21)$$

Here, each query $\mathbf{q}_t$ learns to retrieve from $\mathbf{U}$ the information needed to reconstruct the t -th acoustic frame. This design turns Mel reconstruction into a sequence-to-sequence generation problem from token-aligned text-prosody representations to frame-aligned acoustic features.

Finally, let $\mathbf{D} = (\mathbf{d}_1, \dots, \mathbf{d}_L)$, where $\mathbf{d}_t \in \mathbb{R}^{d_r}$ denotes the decoder output corresponding to the t -th Mel frame query. Each decoder output vector is then mapped to a Mel frame through a multilayer projection head:

$$\hat{\mathbf{m}}_t = g(\mathbf{d}_t) \in \mathbb{R}^F, \quad (22)$$

where $g(\cdot)$ is a feed-forward network and $F = 128$. Collecting all reconstructed frames gives the predicted Mel spectrogram

$$\hat{\mathbf{M}} = (\hat{\mathbf{m}}_1, \dots, \hat{\mathbf{m}}_L) \in \mathbb{R}^{F \times L}. \quad (23)$$

Overall, the full computation pipeline can be summarized as

$$\mathbf{x} \rightarrow \mathbf{M} \rightarrow \mathbf{H}^{\text{enc}} \rightarrow \{\mathbf{y}, \mathbf{P}\} \rightarrow \mathbf{Z} \rightarrow \mathbf{U} \rightarrow \mathbf{D} \rightarrow \hat{\mathbf{M}}. \quad (24)$$

This formulation explicitly decomposes spoken input into a semantic stream and a prosodic stream, and then requires the model to jointly use both streams to reconstruct the original acoustics.

C Implementations Details of the Probing Experiments

For all probing experiments, we train lightweight MLP classifiers on top of the extracted prosody embeddings. We evaluate both a 1-layer and a 2-layer probe. The 1-layer probe consists of a dropout layer followed by a linear classifier, while the 2-layer probe consists of a linear layer, a ReLU activation, a dropout layer, and a final linear classifier. For the 2-layer probe, the hidden dimension is set to 256. Across all probing experiments, we use the Adam optimizer with learning rate 10^{-3} , dropout rate 0.1, batch size

Table 6: Number of samples in each label of our accent benchmark.

Target label	Samples
United States English	310
England English	310
India and South Asia (India, Pakistan, Sri Lanka)	309
Europe	71
Total	1,000

Table 7: Common Voice accent strings grouped into the “Europe” category in our accent benchmark.

Common Voice labels mapped to “Europe”
German; French; Russian; German English; Dutch; Icelandic; Dutch English; Polish; German Accent; French Accent; Slovak; Deutsch English; English with a French accent; German native; native Dutch speaking; Swedish; Finnish

32, and cross-entropy loss. The maximum number of training epochs is set to 20. We apply early stopping based on validation accuracy with a patience of 3 epochs, and the checkpoint with the best validation accuracy is used for the final evaluation.

For the evaluation protocol, we do not rely on a single train-test split. Instead, we adopt a stratified multi-seed cross-validation setup. Specifically, for each probe architecture, we run 10 random seeds and, under each seed, perform 5-fold stratified cross-validation, resulting in 50 runs in total per probe. In each run, the data split is determined by shuffled stratified folds, and the model is re-initialized independently. We then aggregate the results across all runs. Therefore, the probing numbers reported in the paper are robust estimates obtained from repeated stratified cross-validation.

D Accent Benchmark Construction Details

We construct the accent benchmark from the English split test set of Common Voice 22.0. by selecting four target labels that are both representative of major English accent groups and sufficiently well supported in Common Voice [53]: “United States English”, “England English”, “India and South Asia (India, Pakistan, Sri Lanka)”, and “Europe”. Following AIR-Bench [61], the final benchmark contains 1,000 samples, with the per-category distribution summarized in Table 6. Because Common Voice does not provide a single label named “Europe”, we map multiple Europe-related accent strings to this category. All the mapped strings are summarized in Table 7.

E WhisperPro Design Choice Investigations

We study two central design choices in WhisperPro. **Prosody layer.** We vary which Whisper decoder layer is used to extract the prosody embedding and test representative early, middle, and late layers (layers 4, 16, and 30 out of 32 decoder layers). The reconstruction is trained on the prosody embeddings produced by the corresponding layer. **Loss ratio (λ).** We vary the weight that balances the ASR loss and the reconstruction loss, considering $\lambda \in \{0.5, 1.0, 2.0\}$. We evaluate each setting on both a semantic task and a prosodic task. For semantic evaluation, we pair the resulting encoder with Qwen2.5-7B-Instruct and measure question-answering accuracy on OpenBookQA. For prosodic evaluation, we use the 2-layer MLP probing setup on SER. As shown in Figure 3, performance is broadly stable across these choices, suggesting that WhisperPro is robust to moderate changes in both the selected prosody layer and the loss ratio.

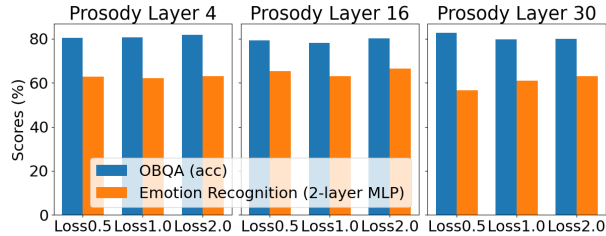


Figure 3: Design choice investigation results for WhisperPro. The blue and orange bars represent OpenBookQA accuracy and the 2-layer MLP probing results on the SER task.

F LLM Design Choice Investigations

We investigate the effect of two critical design choices in LLM training.

1) Knowledge Distillation Loss Type: In prior studies that address the modality gap problem, they typically use Kullback-Leibler (KL)-divergence loss for knowledge distillation [11, 31]. Especially, SALAD [11] finds that KL loss achieves significantly better modality alignment results. Therefore,

we investigate whether KL loss or cross entropy loss provides better results. Specifically, we use a token-level forward KL objective in which the teacher and student are conditioned on different input sequences:

$$\mathcal{L}_{\text{KL}} = T^2 \sum_{j \in \mathcal{A}} \text{KL} \left(P_t \left(\cdot \mid \mathbf{x}_{\leq j}^{(t)} \right) \parallel P_s \left(\cdot \mid \mathbf{x}_{\leq j}^{(s)} \right) \right), \quad (25)$$

where $\mathcal{A}$ denotes the set of answer-token positions used for distillation, $\mathbf{x}^{(t)}$ and $\mathbf{x}^{(s)}$ denote the teacher and student input sequences, respectively, and T is the distillation temperature. In our setting, the teacher input $\mathbf{x}^{(t)}$ is the clean text-only sequence, while the student input $\mathbf{x}^{(s)}$ is the corresponding sequence containing ASR text tokens and its injected prosody embedding. The predictive distributions are obtained from the teacher and student logits as

$$P_t \left(\cdot \mid \mathbf{x}_{\leq j}^{(t)} \right) = \text{softmax} \left(\frac{\mathbf{z}_j^{(t)}}{T} \right), \quad P_s \left(\cdot \mid \mathbf{x}_{\leq j}^{(s)} \right) = \text{softmax} \left(\frac{\mathbf{z}_j^{(s)}}{T} \right), \quad (26)$$

where $\mathbf{z}_j^{(t)}$ and $\mathbf{z}_j^{(s)}$ are the teacher and student logits at position j . This formulation makes explicit that we distill the teacher’s next-token distribution into the student’s next-token distribution under different conditioning contexts.

In addition to the KL term, our training objective also includes the standard next-token cross-entropy loss. We balance the two losses using a coefficient $\alpha \in [0, 1]$, yielding the overall objective

$$\mathcal{L} = \alpha \mathcal{L}_{\text{KL}} + (1 - \alpha) \mathcal{L}_{\text{CE}}, \quad (27)$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss on the target tokens and $\mathcal{L}_{\text{KL}}$ is the distillation loss defined above. To study the effect of this trade-off, we evaluate $\alpha \in \{0, 0.5, 1\}$. Here, $\alpha = 0$ corresponds to pure cross-entropy training, $\alpha = 1$ corresponds to pure KL distillation, and $\alpha = 0.5$ gives equal weight to the two objectives. For all distillation experiments, we fix the temperature to $T = 2.0$.

2) Knowledge Distillation Text Token Source: Because WhisperPro internally performs ASR to produce text tokens, an important design question is which text source should be used during distillation. More specifically, we ask whether the student should be trained with ground-truth text, with ASR-transcribed text, or with a mixture in which the student input and teacher target are derived from different text sources. To answer this question, we compare the following three settings:

1. **GTQ_GTA:** The student input uses the original ground-truth text, and the teacher distribution is also computed from the original ground-truth text. This is the cleanest setting, and it corresponds to distillation under an idealized assumption of error-free transcription.
2. **ASRQ_ASRA:** The student input uses the text transcribed by WhisperPro from the speech-form data, and the teacher distribution is also conditioned on this transcribed text. This setting matches inference-time conditions most closely, because the student is trained on the same type of text tokens it will receive at test time. However, the distillation target is also based on text that may contain ASR errors.
3. **ASRQ_GTA:** The student input uses the text transcribed by WhisperPro, while the teacher distribution is computed from the original ground-truth text. This setting exposes the student to realistic ASR-imperfect inputs, but still provides a clean teacher target. Intuitively, it tests whether the model can learn to recover from typical ASR errors by mapping noisy transcribed inputs toward the behavior induced by the correct text.

For all three settings, the student always receives the same prosody embeddings in addition to the text tokens during training. Additionally, we use only the knowledge distillation datasets (CommonsenseQA and UltraChat) for training. We evaluate all settings on MMSU and OBQA benchmarks. Regardless of the text source used during training, all text tokens at inference time are produced by WhisperPro. As shown in Table 8, we do not observe large performance differences across the three settings. However, the ASRQ_ASRA setting with $\alpha = 0$ achieves the best overall results across the two benchmarks, so we adopt this configuration in the final model training.

G More Related Works

Several prior works also factorize or decouple different aspects of speech, but they are largely motivated by *generation quality*, *prosody-rich synthesis*, or *prosody control*, rather than by reducing

Table 8: Effect of loss weight α and text-token source on MMSU and OBQA (%).

Setting	α	MMSU	OBQA
GTQ_GTA	0.0	67.1	82.9
GTQ_GTA	0.5	63.0	80.9
GTQ_GTA	1.0	66.5	82.2
ASRQ_ASRA	0.0	68.0	84.4
ASRQ_ASRA	0.5	66.5	81.5
ASRQ_ASRA	1.0	63.0	79.8
ASRQ_GTA	0.0	67.6	82.4
ASRQ_GTA	0.5	65.0	81.1
ASRQ_GTA	1.0	63.7	81.1

the semantic modality gap between a TLM and an SLM. For example, AudioLM [39], pGSLM [68], SpiritLM [15], Moshi [3], and SpeechGPT-Gen [69] separate semantic/content information from lower-level acoustic, pitch, style, or codec streams so that the model can better generate natural and expressive speech. A related line of work focuses more explicitly on controllable generation. NaturalSpeech 3 [70] factorizes speech into multiple codec subspaces such as content and prosody and then generates them with a cascade of diffusion modules, while ProsodyLM [71] converts each utterance into text together with discrete word-level prosody tokens so that an LLM can better model prosodic phenomena. These approaches show the value of disentangling content from other speech attributes, but their main objective is still to improve spoken generation or prosody control.

Our motivation is different. We are not primarily trying to make the decoder produce more expressive speech; instead, we seek to make spoken *input* look more like the input of a text LLM while preserving paralinguistic information needed for understanding. Concretely, TextPro-SLM uses synchronized text tokens and compact continuous prosody embeddings, so the LLM can retain the semantic strengths of the original TLM while gaining stronger paralinguistic understanding. This differs from prior token-factorization approaches that represent prosody or acoustics as additional discrete streams or interleaved tokens, which are well suited for generation but can substantially lengthen the sequence and shift the interaction format away from standard TLM processing. In this sense, our method is closer to an input-side alignment strategy: the goal is better semantic preservation under speech input, rather than richer speech synthesis alone.

H Broader Societal Impact

Our work studies how to reduce the modality gap in SLMs by improving the input-side speech representation rather than relying only on output-side adaptation. A potential positive societal impact of this direction is that it may make speech-based AI systems substantially more semantically capable, allowing users to access stronger reasoning, knowledge use, and instruction-following abilities directly through spoken interaction. In settings such as voice assistants, spoken question answering, accessibility tools, and hands-free interfaces, reducing the modality gap could make speech systems more reliable for general tasks, especially when current models lose important meaning when moving from text input to speech input.

At the same time, making SLMs more semantically powerful can introduce meaningful risks. First, if a model becomes better at extracting task-relevant meaning from speech, users may place greater trust in spoken interfaces even when they still make subtle but consequential mistakes in factual understanding, reasoning, or instruction interpretation; such failures could be especially harmful in high-stakes domains. Second, more capable speech interfaces may accelerate deployment of automated systems in education, customer service, or other communication-heavy settings, potentially amplifying harms from misinformation, over-reliance, or reduced human oversight on a larger scale. Third, although not the primary focus of our work, better use of speech-side cues such as prosody may also increase the possibility that systems infer sensitive speaker attributes or intentions in ways users do not expect, raising privacy and fairness concerns if such capabilities are deployed carelessly.